DATE: November 2026

CURRICULUM VITAE

Byeong Kil Lee

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Research Interests (keywords)

Computer architecture, AI acceleration - hardware and software, deep-learning-based intelligent computing (AI computing), application-specific embedded systems (mobile processors), diffusion applications (characterization, acceleration), data augmentation (MRI data generation for AD, MS), memory performance (last-level cache, TLB, HBM), workload characterization of emerging applications, data analysis, parallel computing and parallel architecture design, performance modeling, low power design, early-stage power estimation, and FPGA prototyping.

Education

- **Ph.D.** in Computer Engineering, University of Texas at Austin, August 2005
  Dissertation: Network Processor Design: Benchmarks and Architectural Alternatives (Advisor: Lizy John)
- **M.S**. in Electrical Engineering, KyungPook National University, Korea 1991
- **B.S.** in Electrical Engineering, KyungPook National University, Korea 1989

Professional Experience

Full-time position

- **August 2018 – present, *Associate Professor*, University of Colorado, Colorado Springs**
  -Research, teaching, and services
  -Teaching courses: Logic Circuits II, Computer Architecture and Design, Deep Learning
  -Supervising 2 Ph.D. students, 2 master students, and 3 undergraduate students in my research lab, LiCA (Lab for Computer Architecture)

- **June 2013 – December 2017, *Vice President (Research)*, Samsung**
  -Managed a worldwide project for mobile multimedia HW/SW IP development focused on accelerating Image/video solutions, AR/VR solutions, and Deep learning solutions (Samsung Galaxy 5/6/7/8/9)
  -Research and development on mobile AP hardware/software optimization, ESL simulation, mobile DSP architecture design, Performance modeling, Machine learning HW/SW solutions, etc.
  -**Adjunct Professor (March 2015 – February 2017), Sungkyunkwan University (SKKU)**
  .Samsung & University program (teaching classes at two universities – SKKU & Korea University)

- **August 2009 – June 2013, *Assistant Professor*, The University of Texas at San Antonio**
  -Supervised 2 Ph.D. students, 13 master students, and 30 undergraduate students in my research lab, CASL (Computer Architecture and Systems Lab)
  -Teaching courses: Computer Architecture (G), Computer performance evaluation and Benchmarking (G), Digital systems design (UG), Computer Organization (UG)
  -Managed NSF-funded REU Site (Undergraduate summer research program, ESCAPE: Experimental Study on Computer Architecture and Performance Evaluation) as a PI, supervised 30 students for 10 weeks each summer
  -Published a book on "Digital Systems Design using Verilog", coauthored with Charles H. Roth and Lizy Kurian John. (Publishing company: Cengage Learning)
  -Served as several international conference/Journal committee members and reviewer

- **July 2004 –August 2009, *Senior Design Engineer*, Texas Instruments (Austin)**
  -October 2008– August 2009, RTL design for ARM Cortex A9 dual-core wireless embedded subsystem
  -August 2007–September 2008, RTL design for a DSP system for wireless embedded platform – OMAP4
  -July 2004– July 2007, ARM CortexA8 implementation for wireless embedded system – OMAP3

- **February 1991 – June 2000, *Senior Research Staff*, Agency for Defense Development (Korea)**
  -Embedded system/software development for GIS/GPS military navigation system
  -Geographic information system design using X-window (motif), C/C++, and Oracle DB-SQL
  -Image transmission/processing system design and its software development

Part-time position

- **June 2000 – July 2004 (4 yrs), Research Assistant / Teaching Assistant, UT-Austin (ECE)**
  (Projects involved: DARPA, IBM, NSF, Texas Instruments, and Motorola projects)
  -Development of a benchmark suite for network processors and its workload characterization
  -Implications of executing compression and encryption applications on general-purpose processors
  -Hardware acceleration for application-specific processor
  -Assisted with teaching undergraduate electrical and computer engineering classes

Publications

Book and Book Chapters

1. (Anthony P. Ambler, John W. Sheppard) Byeong Kil Lee, Realizing Complex Integrated Systems - Chapter 6: Performance Evaluation, CRC Press, February 2025.
2. Charles H. Roth, Lizy Kurian John and Byeong Kil Lee, Digital Systems Design using Verilog, Publishing company: Cengage Learning, 2015

Journal Papers

1. S. Anik, J. Ryoo, B. Lee, "PREDICaT: Prefetch-Aware Reuse Distance Cache Tracker Leveraging LLC Access Patterns," IEEE Access, 2026. (under review)
2. S. Anik, B. Lee, J. Ryoo, "Layer-Matched Performance Analysis of Optimized 3D Diffusion Models across GPU Generations," ACM Transactions on Modeling and Performance Evaluation of Computing Systems (ToMPECS), 2026. (under review)
3. S. Lee, B. Lee, S. Seo, "SwitchDetect: Detecting Retrieval-Augmented AI-Generated Text via Context Switching and Token Probability Shifts," IEEE Access, 2026. (under review)
4. J. J. Malone and B. K. Lee, "Performance and Computational Efficiency of LMS Adaptive Volterra Equalizers for Nonlinear TWTA Distortion in Satellite Communications," in IEEE Access, vol. 12, pp. 149740-149759, 2024
5. S. Anik, K. Kelso, B. Lee, "Efficient Layer Optimizations for Deep Neural Networks," International Journal of Soft Computing and Engineering, 2024
6. Lily Young, James Richardson York, and Byeong Kil Lee, "Implications of Deep Compression with Complex Neural Networks," ***International Journal of Soft Computing and Engineering (IJSCE)***, vol. 13, Issue 3, pp.1-6, July 2023
7. J. Pattee, S. M. Anik, and Byeong Kil Lee, "***Performance Monitoring Counter Based Intelligent Malware Detection and Design Alternatives***," in IEEE Access, vol. 10, pp. 28685-28692, 2022.
8. Byeong Kil Lee, "Workload Pruning for Effective Architecture Exploration," ***International Journal of Soft Computing and Engineering (IJSCE)***, vol.8, Issue 6, pp. 7-17, March 2019.
9. Ho Young Kim, Robert Maxwell, Ankil Patel, and Byeong Kil Lee, "Performance Evaluation of Multi-Threaded System vs. Chip-Multi-Processor System," ***International Journal of Computer Technology and Electronics Engineering (IJCTEE)***, vol.2, Issue 4, pp.10-17, August 2012

10. Fahian Ahmed, Saddam Quirem, Gak Min and Byeong Kil Lee, "Hotspot Analysis Based Partial CUDA Acceleration of HMMER 3.0 on GPGPUs," ***International Journal of Soft Computing and Engineering (IJSCE)***, vol.2, Issue 4, pp.91-95, September 2012
11. Satish Raghunath, Lakshmi Deepika Bobbala, Naveen Davanam, and Byeong Kil Lee, "Divide-and-Conquer Way Access for Low Power Mobile Caches**," *International Journal of Computer and Electrical Engineering (IJCEE)***, vol.3, No.2, pp.297-302, 2011
12. Lakshmi Deepika Bobbala, Monobrata Debnath and Byeong Kil Lee, "Composite Pseudo Associative Cache with Victim Cache for Mobile Processors**," *Journal of Computer Science*** 7 (10): pp. 1448-1457, 2011
13. Byeong Kil Lee, "Exploiting Statically Identified ILP for Network Processor Applications," ***International Journal of Computer and Electrical Engineering (IJCEE)***, vol. 2, no. 5, pp. 830-837, October 2010
14. Byeong Kil Lee and Lizy Kurian John, "Hardware Acceleration for Media/Transaction Applications in Network Processors," ***IEEE Transactions on Very Large Scale Integration (VLSI) Systems,*** vol. 17, no. 12, pp.1691-1697, December 2009
15. Byeong Kil Lee, Lizy Kurian John, and Eugene John, "Architectural Enhancements for Network Congestion Control Applications," ***IEEE Transactions on Very Large Scale Integration (VLSI) Systems***, vol. 14, pp. 609- 615, 2006
16. Byeong Kil Lee, Lizy Kurian John, "Implications of Executing Compression/Encryption Applications on General Purpose Processors," ***IEEE Transactions on Computers***, vol. 54, pp. 917-922, 2005
17. Byeong Kil Lee, Sangzee Lee, "A Study on Perspective Display Using 3D Elevation Data with 2D Information Overlay," ***Journal of The Korean Society of Broadcast Engineers***, vol. 2, pp. 36-43, 1997
18. Byeong Kil Lee, Sangzee Lee, "A Study on the GIS Feature and Attribute Coding Scheme," ***Journal of The Korean Society for Geo-Spatial Information System***, vol. 3, pp. 63-74, 1995
19. Byeong Kil Lee, Soonja Kim, Youngho Ha, "Space-variant B-spline functions for Image Interpolation," ***The Transactions of Korean Institute of Electrical Engineers***, vol. 40, pp. 394-401, 1991.

Conference Papers

1. Dazhi Yang, Shafayat Mowla Anik, Byeong Kil Lee, Jeeho Ryoo, "Forecast-Referenced Exact Differential Execution for Diffusion Transformers," ISCA 2027 (to be submitted)
2. Shafayat Mowla Anik, Dazhi Yang, Jeeho Ryoo, Byeong Kil Lee, "Architecture- and OS-Level Characterization of SPEC CPU2026 and CPU2017 on a Production HPC System," ISPASS 2027 (to be submitted)
3. T. Ha, S. Anik, D. Lee, B. Lee, J. Ryoo, "Temporal-Attention Head Specialization During Video Diffusion Training," ICLR 2027 (under review)
4. J. Ryoo, Y. Jung, M. Khaliq, W. Zhang, B. Lee, "Performance Analysis and Optimization of 3D Generative Diffusion Models across GPU Architectures," IEEE International Conference on Performance Engineering (ICPE), May 2026 ***Best paper award nomination***
5. Dazhi Yang, Shafayat Mowla Anik, Byeong Kil Lee, Jeeho Ryoo, "Activation Concentration: Characterizing Column-Level Output Sparsity Across Diffusion Model Architectures," IEEE International Symposium on Workload Characterization (IISWC), 2026
6. Y. Jung, S. Anik, B. Lee, J. Ryoo, "RH+: Row-Hit-Optimized Scheduling for PIM-based LLM Inference," IEEE International Conference on Application-specific Systems, Architectures, and Processors (ASAP), 2026
7. W. Hileman, B. Lee, M. Trimboli, G. Plett, "A Physics-Informed Neural-Network Model for Rechargeable Thick-Electrode Lithium-Metal Battery Cells that is Accurate at High C-Rate," American Control Conference (ACC), 2026
8. W. Zhang, Y. Jung, S. Anik, F Xiao, E. Chuluunbaatar, B. Lee, and J. Ryoo, "Lesion-DDPM: Lesion-Enhanced 3D Diffusion for MS MRI Synthesis," IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR), 2026
9. Shafayat Mowla Anik, Yongchan Jung, Jeeho Ryoo, Byeong Kil Lee, "Dead on Arrival: Characterizing and Protecting Against Dead-Entry TLB Misses in GPU Microarchitectures," IEEE International Performance Computing and Communications Conference (IPCCC), 2026

10. Muge Zhang, Muhammad Ali Khaliq, Jamal Alsakran, Byeong Kil Lee and Jeeho Ryoo, "Structural MRI Synthesis for Alzheimer's Disease via Conditional Diffusion on Anatomical Masks," IEEE International Conference on Multimedia Information Processing and Retrieval (MIPR), 2025
11. Daniel Miller and Byeong Kil Lee, "Appearance-Based Gaze Estimation with Deep Learning for Interactive Digital Signage", International Conference on Image Processing, Computer Vision & Pattern Recognition, 2025
12. Adam Smith, Judah Small, and Byeong Kil Lee, "Implications for Designing Hawks Detection with Data Augmentation and Network Optimizations," The 28th International Conference on Image Processing, Computer Vision, & Pattern Recognition, July 2024
13. Caleb Hill and Byeong Kil Lee, "Classification of Radio Signal Modulations using Convolutional Neural Network," The 23rd International Conference on Wireless Networks, July 2024
14. Shafayat Anik and Byeong Kil Lee, "Access Pattern Characterization of Last-level Cache for Effective Replacement," International Conference on Computational Science and Computational Intelligence, December 2023
15. Joe Tran and Byeong Kil Lee, "EFI: Cache Replacement Policy using Eviction Frequency Integration," The 21st International Conference on Embedded Systems, Cyber-physical Systems, & Applications, July 2023
16. Lily Young and Byeong Kil Lee, "A Reinforcement Learning Approach to Training Chess Engine Neural Networks," The 25th International Conference on Artificial Intelligence, July 2023
17. Jared Smit, Byeong Kil Lee, "Automatic Music Mastering using Deep Learning," 9th Annual Conf. on Computational Science & Computational Intelligence, December 2022
18. Shafayat Anik and Byeong Kil Lee, "Trace Characterization based Cache Replacement Policy," IEEE International Performance Computing and Communications Conference (IPCCC), November 2022
19. Kevyn Kelso and Byeong Kil Lee, "Accordion Autoencoders (A^2E) for Generative Classification with Low Complexity Network," International Symposium on Computational Intelligence, December 2021
20. Tristan Lee, Byeong Kil Lee, "Splatter: An Efficient Sparse Image Convolution for Deep Neural Networks," International Symposium on Computational Intelligence, December 2021
21. J. Chism and B. Lee, "Performance Characterization of Supervision on Knowledge Distillation," Future Technologies Conference, 2021
22. Marina Villalba Carballo and Byeong Kil Lee, "Accuracy-aware Structured Filter Pruning for Deep Neural Networks," The 2020 CSCI - International Symposium on Artificial Intelligence, December 2020
23. Autumn Knight and Byeong Kil Lee, "Performance Analysis of Network Pruning for Deep Learning based Age-Gender Estimation," The 2020 CSCI - International Symposium on Signal & Image Processing, Computer Vision & Pattern Recognition, December 2020
24. Jordan Pattee and Byeong Kil Lee, "Design Alternatives for Performance Monitoring Counter based Malware Detection," IEEE International Performance Computing and Communication Conference, November 2020
25. Jordan Pattee and Byeong Kil Lee, "Intelli-dynamic Malware Detection Based on Processor Behaviors," ITNG 11th International Symposium on Cybersecurity, April 2020
26. Jordan Pattee and Byeong Kil Lee, "Implications for hardware Acceleration of Malware Detection," IEEE International Conference on Application-specific Systems, Architectures, and Processors (ASAP), July 2019
27. Jordan Pattee and Byeong Kil Lee, "Feature Characterization Based on Processor Behaviors of Malwares," International Conference on Data Intelligence and Security (ICDIS), June 2019 ***Best poster award***
28. William Ross and Byeong Kil Lee, "Effective Data Placement to Reduce Cache Thrashing in Last Level Cache," International Conference on Information Technology New Generation (ITNG 2019) - High-Performance Computer Architecture track, April 2019
29. Jungsoo Kim, Hansu Cho, Seungwook Lee, Byeong Kil Lee, Parisa Razaghi, Xinnian Zheng and Andreas Gerstlauer, "A Scalable Simulation Framework for Network/System Co-Simulation," Design Automation Conference (DAC), 2015

30. Nilufar Ferdous, Monobrata Debnath, Byeong Kil Lee, and Eugene John, "Exploiting Reuse-Frequency with Speculative and Dynamic Updates in an Enhanced Directory Based Coherence Protocol," International Conference on Parallel and Distributed Processing Techniques and Applications, 2014
31. Nilufar Ferdous, Byeong Kil Lee and Eugene John, "Performance Enhancement in Shared-Memory Multiprocessors using Dynamically Classified Sharing Information," 33rd IEEE International Performance Computing and Communication Conference (IPCCC), 2014
32. Kathlene Hurt and Byeong Kil Lee, "Proposed Enhancements to Fixed Segmented LRU Cache Replacement Policy," 32nd IEEE International Performance Computing and Communication Conference (IPCCC), 2013
33. Jie Luo, Kathlene Morales, Byeong Kil Lee, Eugene John, and Young Kyu Choi, "Performance-Sensitivity and Performance-Similarity Based Workload Reduction," 31st IEEE International Performance Computing and Communications Conference (IPCCC), December 2012
34. Saddam Quirem and Byeong Kil Lee, "AES Decryption Using Warp-Synchronous Programming," 31st IEEE International Performance Computing and Communications Conference (IEEE IPCCC), December 2012
35. Kathlene Hurt and Byeong Kil Lee, "Fixed Segmented LRU Cache Replacement Scheme with Selective Caching," 31st IEEE International Performance Computing and Communications Conference (IPCCC), December 2012
36. Fahian Ahmed, Byeong Kil Lee, Bum Joo Shin, Duk Soo Son, Young Choon Woo, and Wan Choi, "Performance Hotspot Based CUDA Acceleration," International SoC Design Conference (ISOCC 2012), November 2012
37. Savithra Eratne, Eugene John, Byeong Kil Lee, "Reducing Thermal Hotspots in Microprocessors with Expanded Component Sizing," International Midwest Symposium on Circuits and Systems, 2012
38. Jie Luo, Yilin Zhang, Saarvaani Vadlamani and Byeong Kil Lee, "Performance-Sensitivity-based Workload Tailoring for Effective Design Exploration," 9th International Conference on Information Technology New Generation (ITNG 2012) - High-Performance Computer Architecture track, April 2012
39. Fahian Ahmed, Saddam Quirem, Bumjoo Shin, Youngchoon Woo, Dukjoo Son and Byeong Kil Lee, "A Study of CUDA Acceleration and Impact of Data Transfer Overhead in Heterogeneous Environment," International Workshop on Unique Chips and Systems (UCAS-7), 2012
40. Monobrata Debnath, Ankil Patel, and Byeong Kil Lee, "A Study on Performance Impact of Network Delay in Chip Multi Processors," International Workshop on Unique Chips and Systems (UCAS-7), 2012
41. Saddam Quirem, Fahian Ahmed and Byeong Kil Lee, "CUDA Acceleration of P7Viterbi Algorithm in HMMER 3.0," The 30th IEEE. International Performance Computing and Communications Conference (IPCCC 2011), November 2011
42. Naveen Davanam, Ho Young Kim, Byeong Kil Lee, and Eugene John, "Sensitivity Analysis on Hardware Resources in SMT Processors," The 2011 International Conference on Computer Design (CDES'11), 2011
43. Claudia Romo, Savithra Eratne, Eugene John, and Byeong Kil Lee, "Leakage - Delay Tradeoff in Wide-Bit Nanoscale CMOS Adders," The 2011 International Conference on Computer Design (CDES'11), 2011
44. Bobbala Lakshmi Deepika and Byeong Kil Lee, "Hybrid-way Cache for Mobile Processors," 8th International Conference on Information Technology New Generation (ITNG 2011) - High-Performance Computer Architecture track, April 2011
45. Satish Raghunath and Byeong Kil Lee, "Selection of Representative Simulation Point using Performance Metric based Similarity," Sixth Workshop on Unique Chips and System (UCAS-6), December 2010
46. Satish Raghunath and Byeong Kil Lee, "Effective Workload Reduction for Early-stage Power Estimation," International SoC Design Conference (ISOCC 2010), November 2010
47. Satish Raghunath, Naveen Davanam, Bobbala Lakshmi Deepika and Byeong Kil Lee, "Way-Load Balancing Scheme for Mobile Cache LRU Replacement," The 23rd IEEE International SOC Conference (IEEE SOCC 2010), September 2010
48. Bobbala Lakshmi Deepika, Javier Salvatierra, and Byeong Kil Lee, "Composite Pseudo-Associative Cache for Mobile Processors," The 18th IEEE International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems (MASCOTS 2010), August 2010

49. Byeong Kil Lee, Satish Raghunath, and Eugene John, "Architectural Sensitivity Analysis on Network Workloads," The 2010 International Conference on Computer Design (CDES'10), July 2010
50. Savithra Eratne, Claudia Romo, Eugene John, and Byeong Kil Lee, "Leakage and Access Time Tradeoffs for Cache in High-Performance Microprocessors," The 2010 International Conference on Computer Design (CDES'10), July 2010
51. Naveen Davanam and Byeong Kil Lee, "Towards Smaller-sized Cache for Mobile Pro-cessors using Shared Set-associativity," 7th International Conference on Information Technology New Generation (ITNG 2010) - High-Performance Computer Architecture track, Las Vegas, April 2010
52. Monobrata Debnath, Byeong Kil Lee, and Wei-Ming Lin, "Prioritized Out-of-Order Instruction Dispatching Techniques for Simultaneous Multi-Threading (SMT) Processors," IEEE Real-Time Systems Symposium (RTSS '09), December 2009
53. Byeong Kil Lee and Bhasi Kaithamana, "Verification of Full-custom Blocks using Symbolic Simulation," TI Symposium on Digital Verification, 2007
54. Byeong Kil Lee, Lizy Kurian John, Eugene John, "Architectural Support for Accelerating Congestion Control Applications in Network Processors," IEEE International Conference on Application-specific Systems, Architectures and Processors (ASAP) 2005
55. Byeong Kil Lee, Lizy Kurian John, "NpBench: A Benchmark Suite for Control plane and Data plane Applications of Network Processors," International Conference on Computer Design (ICCD '03), October 2003
56. Byeong Kil Lee, Lizy Kurian John, "Implications of programmable general-purpose processors for compression/encryption applications," 13th IEEE Conference on Application-specific Systems, Architectures and Processors (ASAP), July 2002
57. Byeong Kil Lee, Sangzee Lee, "Advanced Raster Map and Its Application," Proceedings of Geo Spatial Information Age (USA), 1997
58. Byeong Kil Lee, Sangzee Lee, "A Study on Perspective Display Using 3D Elevation Data with 2D Information Overlay," Proceedings of Image Processing Workshop, pp. 281-286, 1997
59. Byeong Kil Lee, Sangzee Lee, "Compression of the Vector Restored Raster Graphics(VRRG)," Proceedings of Image Processing Workshop, 1995.
60. Byeong Kil Lee, Sangzee Lee, "A Study on Ray-tracing Approach to Perspective Image Display," Proceedings of The Korean Signal Processing Conference, pp. 81-86, 1994
61. Byeong Kil Lee, Sangzee Lee, "A Study on Self-Organized Distributed Networks for Terrain Elevation Data Compression and Storage," Proceedings of The Korean Institute of Telematics and Electronics Conference, pp. 569-1572, 1994
62. Byeong Kil Lee, Sangzee Lee, "A Study on the Comparison of the Digitized Raster Map Using Vector Quantizer," Proceedings of The Korean Signal Processing Conference, pp. 179-183, 1994
63. Byeong Kil Lee, Youngho Ha, "Locally adaptive Image resampling using new spline functions," Proceedings of The Korean Signal Processing Conference, pp. 370-374, 1990

Non-Referred Publications

Technical Reports

1. Byeong Kil Lee, Lizy Kurian John, "Development and Characterization of Control-plane Network Workloads," TR-030827-01, ***UT-Austin***, 2003
2. Byeong Kil Lee, Lizy Kurian John, "Memory characteristics of Compression/Encryption Applications," TR-020315, ***UT-Austin***, 2002
3. Byeong Kil Lee, Lizy Kurian John, "Access Time and Power Characteristics of Various Future File Configurations," TR-020821-01, ***UT-Austin***, 2002
4. Byeong Kil Lee, Sangzee Lee, "A Report on Design and Display Technique for Situation Display," ***ADD Technical Report***, GEDC-509-990932, 1999.
5. Byeong Kil Lee, Sangzee Lee, "A Study on Situation and Geographic Information System," ***ADD Technical Report***, CESD-509-970664, 1997.

Grants and Scholarship

(Pending proposals)

- PI, NIH – "A Deviation-Based Imaging Biomarker for Pseudoprogression Detection in Glioblastoma Using Conditional MRI Synthesis," $325,923 (pending)
- PI, NSF - "Collaborative Research: SHF: Characterization-Driven Power- and Thermal-Aware Scheduling for Memory Hierarchies with Processing-in-Memory," $998,379 (pending)
- PI, NSF - "SHF: Joint Precision and Memory Optimization for Energy Efficient Deep Learning," $673,473 (pending)
- PI, NSF - "ERI: Quantum-Inspired Adaptive Hardware Telemetry for Malware Detection," $199,892 (pending)
- Co-PI, NSF - "S-STEM: Building Institutional Capacity to Support Veterans in STEM Through Scholarships, Transfer Pathways, and Career Preparation" $1,999,027 (pending)
- Co-PI, NSF - "REU Site: Multidisciplinary Engineering Research for Renewable Energy Sustainability at UCCS," $406,838 (pending)

(2026 @UCCS)

- PI, UCCS *internal* – URA (Undergraduate Research Academy), "Architectural Optimization of Sparse Matrix Computation for Efficient Deep Neural Network Processing," $4,000 (May 30, 2026 - August 31, 2026)
- PI, UCCS *internal* – URA (Undergraduate Research Academy), "Performance Bottleneck Analysis of Diffusion Model Execution on GPUs," $4,000 (May 30, 2026 - August 31, 2026)

(2023 @UCCS)

- PI, CU System – "UCCS Faculty Revitalization Fellowship," $20,000 (January 2023 – December 2023)
- PI, UCCS *internal* – URA (Undergraduate Research Academy), "Intelligent Branch Prediction Scheme for Emerging Workloads," $4,000 (May 30, 2023 - August 31, 2023)

(2022 @UCCS)

- PI, UCCS *internal* – URA (Undergraduate Research Academy), "Synthetic Malware Data Augmentation for Deep Learning Network Training," $3,500.00 (May 30, 2022 - August 13, 2022)
- PI, UCCS *internal* – EAS URS (Undergraduate Research Scholars) Program, "Accelerating Sparse Matrix Multiplication for Deep Learning Applications," $3,000 (January 10, 2022 - August 19, 2022).

(2021 @UCCS)

- PI, CU legislative funding, "Reconfigurable Design for PMC-based Malware Detection," $7,500 (May 1, 2021 – November 30, 2021).
- PI, UCCS *internal* – CRCW Seed Grant, "Co-optimization for Deep Learning Neural Networks," $7,500 (May 1, 2021 - August 31, 2022).
- PI, UCCS *internal* – -URA (Undergraduate Research Academy), "Co-design for Deep Learning Accelerators in RISC V Processors," $3,500 (May 1, 2021 - November 30, 2021).
- PI, UCCS *internal* – Provost's office, "A Proposal for Teaching Enhancement Grant," $750 (November 15, 2021 - August 31, 2022).
- PI, UCCS *internal* – Cybersecurity Seed Grant, "Reconfigurable Design for PMC-based Malware Detection," $7,500 (January 2021 - December 2021)

(2020 @UCCS)

- Co-PI, NSF - "MRI: Acquisition of a High-performance Computing Cluster for Next-generation computational Science in Southern Colorado," $621,754 (August 1, 2020 - July 31, 2025).
- PI, UCCS *internal* – Undergraduate Research Academy (URA), "Low Power and Scalability Considerations for Mobile Deep Learning Acceleration," $3,500 (June 2020 - December 2020)
- PI, Symetrix Corporation, "Scalable Architecture Exploration for SNN-based Machine Learning," $40,000 (March 2020)

(2019 @UCCS)

- PI, UCCS *internal* – EAS Undergraduate Research Scholars Program (URS), "Hardware Acceleration of Processor Behavior-based Malware Detection", $4,000 (July 16, 2019 – June 30, 2020)
- PI, UCCS *internal* – Undergraduate Research Academy (URA), "Hybrid approaches for malware detection", $3,500 (June 10, 2019 – August 16, 2019)
- PI, UCCS *internal* – Cybersecurity grant, "Lightweight architectures for machine learning in cybersecurity", $10,000 (January 1, 2019 – July 30, 2019)
- PI, Xilinx (Gift), "Reconfigurable hardware acceleration", FPGA board (Virtex-7) and design suite (2019)

(2018 @UCCS)

- PI, Nvidia (Gift), "Research on exploring novel foundational principles for implementing next generation of AR", Quadro P6000 GPGPU (September 2018)

(2009-2012 @UTSA)

- PI, ETRI, "A Study of Performance Evaluation and Workload Characterization of GPGPU based System", $40,000 (July 1, 2011 - January 31, 2012)
- PI, NSF, "REU Site: ESCAPE: Experimental Study on Computer Architecture and Performance Evaluation", $356,732 (May 1, 2011 – April 30, 2014)
- PI, Wind River (INTEL) - Simics Software package (Gift) (March 15, 2011 - March 15, 2014)
- PI, Apache Design Solutions, Inc.: PowerArtist tool (Gift) (April 1, 2011 - May 31, 2014)
- PI, Nvidia: Academic Partnership with the donation - GPGPU C2075 (April 2012)

Involved Research Projects

- Machine Learning Workload Estimation and Acceleration (Samsung w/ Seoul National University), 2017
  -Motion blur enhancement and tracking solution using machine learning
- Processor Architecture for Mobile Intelligence (Samsung w/ YonSei University), 2017
  -Research on hardware acceleration for machine learning algorithms
- Hardware/Software co-optimization for network system (Samsung w/ KAIST), 2016
  -Hardware acceleration for TCP-IP module on mobile AP
- Performance and Power Optimization for Mobile Devices (Samsung w/ University of Texas at Austin), 2015
  -Scalable simulation framework design for co-simulation
- Web Performance Acceleration (Samsung w/ Sungkyunkwan University), 2015
  -Research on hardware acceleration for mobile web performance
- Low Power SoC for Network Security (Samsung w/ Seoul National University), 2014
  -High-performance and low-power SoC design for network security
- Mobile CPU Profiling for background applications (Samsung w/ YonSei University), 2014
  -Research on diminishing mobile sluggishes
- HMA-based System Optimization (Samsung w/ Seoul National University), 2013-2015
  -System simulation framework & low-power mobile computations
- Network Processor Design: Benchmarks and Architectural Alternatives (NSF), 2003 –2004
  -Parallel accelerator design for network processors
  -Exploiting statically identified parallelism for network processor workloads
- High-performance wireless processor design (Texas Instruments), 2004
  -Machine Description, Instruction-set Simulator Design
  -Physical synthesis, Formal equivalence verification
- Development and Characterization of Control-Plane Network Workloads (Motorola), 2003
  -Development a benchmark suite for network processor design
  -Workload characterization and architectural analysis for emerging network processor application
- TRIPS Project (DARPA), 2001–2002
  -Memory characteristics of multimedia workloads
- C3I Project (ADD Korea), 1991 -2000
  -Development of digital map, geographic information system
  -Software development of 2D/3D map display

Teaching Experience

Courses @UCCS

- ECE2411 (Logic Circuits II, Undergraduate) - Fall 2018, Spring 2019, Spring/Fall 2020, Spring/Fall 2021, Spring/Fall 2022, Spring/Fall 2023, Spring/Fall 2024, Fall 2025, Spring/Fall 2026
- ECE4480/5480 (Computer Architecture and Design, Graduate/Undergraduate)- Spring 2019, Spring 2020, Spring 2021, Spring 2022, Spring 2023, Spring 2024, Spring 2026
- ECE4222/5222 (**Deep Learning**, Graduate/Undergraduate) - Fall 2019, Fall 2020, Fall 2021, Fall 2022, Fall 2023, Fall 2024, Fall 2026
- ECE4450/5450 (Advanced Computer Architecture, Graduate/Undergraduate) - Fall 2019, Fall 2020, Fall 2021, Fall 2022, Fall 2023, Fall 2024, Fall 2026

Courses @UTSA

- EE5123 (Computer Architecture, Graduate) - Spring 2011, Spring 2013
- EE3563 (Digital Systems Design, Undergraduate) - Fall 2009, Fall 2010, Fall 2011, Fall 2012
- EE5423 (Computer Performance Evaluation and Benchmark, Graduate) - Fall 2010, Fall 2012

Recognitions

- CU System – **AI Recognition Award** (AI for Scholarly & Creative Work) 2026
- UCCS College of Engineering and Applied Science (EAS) **Outstanding Teaching Award** 2025
- UCCS College of Engineering and Applied Science (EAS) **Outstanding Teaching Award** 2021
- Selected to receive our Albert Nelson Marquis Lifetime Achievement Award 2019
- Marquis Who's Who in Science and Engineering 2011-2014
- Marquis Who's Who in America 2007-2008
- Outstanding Researcher classification by the US Citizenship and Immigration Services (USCIS) in 2005
- Student Travel Award, College of Engineering, University of Texas at Austin, 2003-2004
- Graduate research assistantship, The University of Texas at Austin (2000 - 2004)
- National Defense Scientific Award (silver medal), Agency Defense Development, Korea (1996)
- Tuition scholarship, Samsung Electronics (1991)
- Tuition scholarship, Department of Electrical Engineering, Kyungpook National University (1985 - 1990)

Presentations at Meetings and Seminars Presented

Invited Presentations

- Co-optimization for Deep Learning Neural Networks, Byeong Kil Lee, UCCS CRCW grant recipient presentation, May 2023
- Synthetic Malware Data Augmentation for Deep Learning Network Training, Sam Peters and Byeong Kil Lee, UCCS Mountain Lion Research Day (MLRD) 2022
- Splatter: An Efficient Sparse Image Convolution for Deep Neural Networks, Tristan Lee and Byeong Kil Lee, UCCS Mountain Lion Research Day (MLRD) 2021
- Accordion AutoEncoders (A^2E) for Generative Classification with Low Complexity Network, Kevyn Kelso and Byeong Kil Lee, UCCS Mountain Lion Research Day (MLRD) 2021
- Performance Analysis of Network Pruning for Deep Learning based Age-Gender Estimation Autumn Knight and Byeong Kil Lee, UCCS Mountain Lion Research Day (MLRD) 2020
- Understanding of AI Neural Network and Deep Learning – Applications and Hardware Accelerators (Korea University of Technology and Education, Korea 2019
- Performance in Mobile SoC, Seoul National University, Korea, 2017
- Processor Architecture for Mobile Intelligence, Yonsei University, Korea, 2016

- Hardware/Software co-optimization for mobile system, Korea University, Korea, 2016
- Performance and Power Optimization for Mobile Devices, 2015 (Sungkyunkwan University)
- Low Power SoC for Mobile Systems, Sungkyunkwan University, Korea, 2015
- SoC Architecture & Early-stage Performance Modeling, Sungkyunkwan University, Korea, 2014
- Mobile AP-SoC Design, Samsung System Architect Forum, 2013
- Mobile Systems Design in Heterogeneous Multi-core Era, Samsung (Suwon), LG (Seoul), and ETRI (Taejeon), Korea, 2011
- Trend of Mobile Processor and Hardware Development for Smart Phone Applications (one week lecture at Korea University of Technology and Education, Korea), 2011
- Microprocessor / SoC Design (one week lecture at Korea University of Technology and Education), 2009
- Design Verification of Full-custom Blocks using Symbolic Simulation, TI, 2007
- Development and Characterization of Control-Plane Network Workloads, Motorola (Austin), 2003
- Characteristics of Media Applications, TRIPS (DARPA) project (UT-Austin), 2001

Professional Organizations

- IEEE Senior Member, 2011 - Present

Services and Leadership

External Activities

- **General chair**
  IEEE Workshop on Unique Chips and Systems (UCAS 2009-2012)
- **Program committee member**
  International Symposium on Computer Architecture (ISCA, 2021, 2023)
  IEEE International Conference on Computer Design - Processor Architecture Track (ICCD, 2009-2013, 2021-2026)
  IEEE International Conference on Parallel Processing – Performance Models Track (ICPP, 2013, 2019-2021)
  International Conference on Application-specific Systems, Architectures and Processors (ASAP, 2019, 2020, 2021)
  IEEE International Performance Computing and Communications Conference (IPCCC, 2018-2026)
  International Conference on Information Technology New Generation (ITNG, 2011-2013, 2018-2021)
  International Conference on Data Intelligence and Security (ICDIS, 2018-2021)
  International Symposium on Cybersecurity (2019, 2020)
  Intelligence Systems Conference (Intellisys, 2019, 2020, 2021, 2022)
  Computing Conference (2019, 2020 - 2024)
  Future Technologies Conference (2021-2024)
  IEEE International Workshop on "Distributed and Intelligent Systems (2021)
  IEEE International Parallel & Distributed Processing Symposium (IPDPS, 2009, 2013)
  IEEE Workshop on Unique Chips and Systems (UCAS, 2005-2012)
  IEEE Workshop on Thermal Modeling and Management (TEMM 2011)
  IEEE International Symposium on Workload Characterization (IISWC 2006)
  IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS, 2006)

- **Organizing committee member**
  IEEE International Conference on Parallel Architectures and Compilation Techniques (PACT 2012, 2020, 2021)
  IEEE International Symposium on High-Performance Computer Architecture (HPCA-17, 2011)
  IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS 2004-2013)

IEEE International Symposium on Workload Characterization (IISWC 2005, 2006, 2008-2010)
IEEE Workshop on Unique Chips and Systems (UCAS, 2005-2011)
IEEE International Workshop on Workload Characterization (WWC, 2001-2004)

- **NSF research proposal review**
  NSF REU proposal review (2012)

- **Session chairs**
  International Conference on Application-specific Systems, Architectures and Processors (ASAP, 2019)
  International SoC Design Conference (ISOCC 2010)
  IEEE Workshop on Unique Chips and Systems (UCAS 2010-2012)

- **Mentoring service to graduate students**
  In cooperation of MentorNet and Texas Instruments, 2007-2009

Reviews of Journal/Conference Papers

<Journal>

IEEE Transaction on Computer (2002, 2005, 2009, 2010, 2019, 2020, 2022, 2023)
IEEE Transaction on Dependable and Secure Computing (2019, 2020, 2021)
ACM Transactions on Architecture and Code Optimization (2005, 2018)
ACM Computing Surveys (2021)
Journal of Cloud Computing (2019, 2020, 2021, 2022)
Journal of Signal Processing Systems (2019, 2020, 2021)
International Journal of Computer and Electrical Engineering (2019)
IEEE Transaction on VLSI (2005, 2010)
IEEE Transaction on Multimedia (2004)
IEEE Transaction on Industrial Electronics (2011)
IEEE Computer Architecture Letters (2011)
International Journal of High-Performance Systems Architecture (IJHPSA, 2009)
Journal of Circuits, Systems, and Computers (JCSC, 2010)
Journal of Low Power Electronics (2010)
International Journal of Advancements in Computer Technology (2011)

<Conference>

Reviewed papers as a committee member (listed on external activities)

(Additionally, some folks asked me to review the paper as a non-Program Committee member)
International Symposium on High-Performance Computer Architecture (HPCA, 2005)
IEEE International Symposium on Workload Characterization (IISWC, 2005)
International Conference on Parallel Architectures and Compilation Techniques (PACT, 2003, 2019)
IEEE/ACM International Symposium on Microarchitecture (2005)
SPEC International Conference on Performance Engineering (SIPEW, 2010)
ACM SIGPLAN Conference on Languages, Compilers and Tools for Embedded Systems (LCTES, 2010)

Status: U.S. Citizen